Rag undertakings

Retrieval Augmented Generation improves LLM responses using external knowledge sources
Model does not depend only on training data
Relevant information is retrieved before generation
System first requires a document collection
These documents act as the knowledge base
Quality of documents directly affects accuracy
Incorrect data produces incorrect answers
Text is divided into smaller chunks
Chunking helps handle model input limits
It also improves retrieval precision
Very large chunks introduce noise
Very small chunks lose important context
Overlap between chunks preserves continuity.
Balanced chunk size gives better meaning retention
Next step Embeddings are vector representations of text
They capture semantic meaning of content
Similar meanings result in similar vectors
Embeddings are stored in vector databases
These databases enable similarity-based search
Distance metrics determine vector closeness
Common vector databases include FAISS and ChromaDB
Pinecone and Weaviate are cloud-based options.
Milvus and Qdrant support large-scale workloads
During a query, user input is embedded
Similarity search retrieves relevant chunks
Retrieved text is given to the language model
Final answer depends heavily on retrieval quality
RAG systems are evaluated using specific metrics which are Faithfulness checks if answer stays within context
Answer relevancy measures alignment with question